

Interpretable Prediction of 30-Day Hospital Readmission in Patients with Diabetes Using Random Forest and SHAP

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Abstract

Unplanned hospital readmission within 30 days of discharge is a persistent source of patient harm and financial penalty for healthcare institutions. Machine learning models can estimate readmission risk, but their opacity limits clinical trust and adoption. This study develops a Random Forest classifier to predict 30-day readmission among patients with diabetes, using the publicly available Diabetes 130-US Hospitals dataset (101,766 encounters, 1999–2008). To avoid dependence between repeated observations, only each patient’s first encounter was retained, and patients who died or were discharged to hospice were excluded, yielding a cohort of 69,973 patients. SHapley Additive exPlanations (SHAP) were used to attribute predictions to individual features. On a held-out test set of 13,995 patients, the model achieved an area under the ROC curve (AUC) of 0.633, with a recall of 0.50 and a precision of 0.14 for the readmitted class, indicating modest discrimination similar to that of a logistic regression baseline (AUC 0.624). SHAP analysis showed that the number of prior inpatient visits, discharge to a skilled nursing facility, the number of medications, and length of stay were the most influential features, with indicators of prior hospital use and clinical complexity ranking above laboratory-related variables. These results suggest that healthcare utilisation history and discharge destination carry predictive information about readmission, and that interpretable models can show clinicians why a patient is flagged. The findings are associative and should not be read as causal.

Keywords: hospital readmission, diabetes, Random Forest, explainable artificial intelligence, SHAP

1. Introduction

Hospital readmission within 30 days of discharge is widely used as a marker of care quality. In the United States, the Hospital Readmissions Reduction Program ties Medicare payments to readmission rates for selected conditions, which gives hospitals a financial incentive to identify at-risk patients before they leave (Jencks et al., 2009). Patients with diabetes are a particularly relevant group because the condition requires continuing management after discharge, including medication adherence, monitoring, and follow-up care.

Predictive models can flag high-risk patients, but a risk score alone is difficult for clinicians to act on. Many high-performing models operate as “black boxes,” and clinicians are understandably reluctant to rely on outputs they cannot inspect. Explainable artificial intelligence (XAI) addresses this by showing which inputs drove a given prediction. SHAP, based on Shapley values from cooperative game theory, provides consistent feature attributions for individual predictions and can be aggregated into global summaries (Lundberg & Lee, 2017; Lundberg et al., 2020).

This study combines a Random Forest classifier (Breiman, 2001) with SHAP to predict 30-day readmission in patients with diabetes and to examine which factors the model relies on. The objectives

were to: (1) build a model with clearly documented cohort selection that avoids repeated-patient dependence; (2) evaluate its predictive performance; and (3) interpret its behaviour using SHAP to identify the most influential features.

2. Methods

2.1 Data source

The analysis used the Diabetes 130-US Hospitals for Years 1999–2008 dataset (Strack et al., 2014), obtained from the UCI Machine Learning Repository. It contains 101,766 inpatient encounters from 130 US hospitals, with demographic variables, admission and discharge information, diagnoses, medications, and the number of prior visits. The outcome variable records whether the patient was readmitted within 30 days, after more than 30 days, or not at all.

2.2 Cohort selection

Many patients appear in the dataset more than once. Treating repeated encounters from the same patient as independent observations violates the independence assumption of standard train–test evaluation and can inflate performance estimates. Encounters were therefore sorted by encounter identifier, used as a proxy for chronological order, and only each patient’s first encounter was retained, leaving 71,518 unique patients.

Patients who died during the index admission or were discharged to hospice care were then excluded (discharge disposition codes 11, 13, 14, 19, 20, and 21), as these patients cannot be readmitted. The final cohort comprised 69,973 patients.

2.3 Outcome and features

The binary outcome was defined as readmission within 30 days (1) versus no readmission or readmission after more than 30 days (0). In the test set, 1,255 of 13,995 patients (9.0%) met the positive definition.

The variables `weight`, `payer_code`, and `medical_specialty` were removed because of extensive missing values. Patient and encounter identifiers were excluded from the predictors. Categorical variables, including the admission type, admission source, and discharge disposition identifiers (which are numeric codes and were converted to categorical type before encoding), were one-hot encoded. Because the diagnosis fields contain many distinct ICD-9 codes, encoding produced 2,291 predictor columns in total. Missing values were not imputed. Entries coded “?” in the dataset (for example in race and the diagnosis fields) were treated as an ordinary category value during one-hot encoding. The A1C result and maximum glucose serum fields are coded “None” when no test was performed (81.6% and 95.2% of patients in the cohort, respectively); these were read as missing and have no indicator column of their own (see Section 4.1).

2.4 Model development and evaluation

The cohort was divided into a training set (55,978 patients) and a test set (13,995 patients) using an 80/20 split, stratified on the outcome, with a fixed random seed (42). A Random Forest classifier was trained with 100 trees, a maximum depth of 10, and balanced class weights to account for the low prevalence of the positive class. No hyperparameter tuning was performed. A logistic regression model with balanced class weights, trained on standardised features from the same training set, served as a baseline. Performance on the held-out test set was assessed using the area under the receiver operating characteristic curve (AUC), together with precision and recall for the readmitted class at the default classification threshold.

2.5 Model interpretation

Feature contributions were computed with the SHAP TreeExplainer (Lundberg et al., 2020) on a random sample of 500 test-set patients, using the contributions toward the readmitted class. Global feature importance was visualised with a SHAP beeswarm plot, in which each point is one patient, horizontal position shows the feature's effect on the predicted risk of readmission, and colour shows the feature's value (for binary indicator variables, red indicates that the patient has the characteristic).

2.6 Software

Analyses were performed in Python 3.12.12 using the Anaconda distribution on a JupyterHub environment, with scikit-learn 1.7.2 (Pedregosa et al., 2011), SHAP 0.52.0, pandas 2.3.3, and matplotlib 3.10.6. Code is available at <https://github.com/katlogakishi033-sys/Readmission-shap>.

3. Results

3.1 Cohort

The cohort selection process reduced 101,766 encounters to 71,518 unique patients and then to 69,973 patients after excluding deaths and hospice discharges. The training set contained 55,978 patients and the test set 13,995.

3.2 Predictive performance

On the test set, the Random Forest achieved an AUC of 0.633. For the readmitted class, precision was 0.14 and recall was 0.50, giving an F1 score of 0.21; for the non-readmitted class, precision was 0.93 and recall was 0.69. Overall accuracy was 0.67. Because only 9.0% of the test patients were readmitted, accuracy is not an informative measure here. The precision of 0.14 for the readmitted class is about 1.6 times the baseline readmission rate, which means that roughly one in seven patients flagged as high-risk was in fact readmitted within 30 days. The model therefore provides modest discrimination, consistent with the general difficulty of readmission prediction. The logistic regression baseline, trained and tested on the same split, achieved an AUC of 0.624, with precision of 0.14 and recall of 0.53 for the readmitted class. The Random Forest's advantage was therefore small (an AUC difference of 0.009) and was not tested for statistical significance.

3.3 Feature importance

Figure 1 shows the SHAP beeswarm plot for the most influential features.

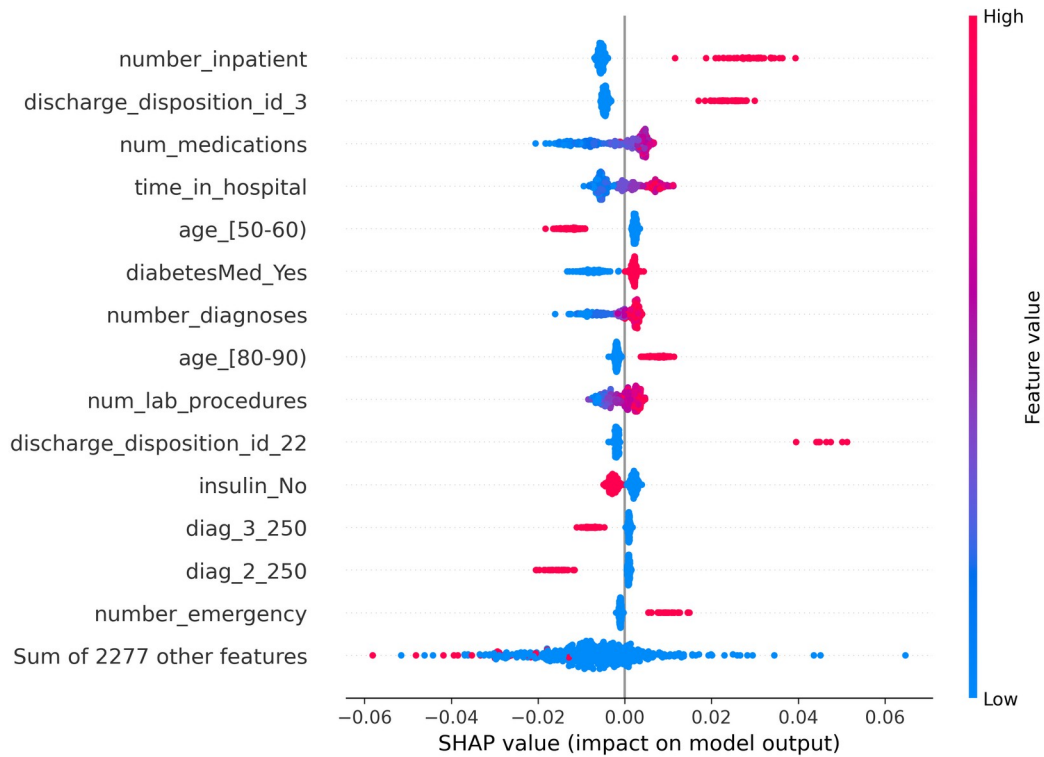


Figure 1. SHAP beeswarm plot of the top features contributing to predicted 30-day readmission risk. Each point represents one patient from a random sample of 500 test patients; colour indicates feature value and horizontal position indicates the effect on the model’s output.

The most influential feature was the number of prior inpatient visits, with higher counts associated with higher predicted readmission risk. The second was discharge to a skilled nursing facility (disposition code 3), which was also associated with higher predicted risk. Further features associated with higher predicted risk were a greater number of medications, longer hospital stays, a greater number of recorded diagnoses, a greater number of prior emergency visits, age 80–90 years, and discharge to a rehabilitation facility (disposition code 22; few patients in the sample). Features associated with lower predicted risk included age 50–60 years, not being prescribed insulin, and the recording of diabetes mellitus (ICD-9 code 250) as a secondary or tertiary diagnosis. Laboratory-related variables, such as the number of laboratory procedures, had smaller effects. Admission type did not appear among the leading features.

4. Discussion

The results suggest that a patient’s history of hospital use, discharge destination, and indicators of clinical complexity carry most of the predictive information in this dataset. Frequent prior inpatient and emergency visits, a higher medication count, and longer stays may reflect patients with more severe or less stable illness, and discharge to a skilled nursing or rehabilitation facility may identify patients who are frailer at the point of discharge. These interpretations are plausible but untested, since SHAP describes what the model relies on and not what causes readmission. The direction of some effects, for example the lower predicted risk for patients with diabetes recorded as a secondary diagnosis, is not obviously clinical and may reflect how diagnoses were coded, so such findings should be treated cautiously.

The comparison between utilisation variables and laboratory variables should also be interpreted with care. Key laboratory measures in this dataset are largely missing, so they had limited opportunity to contribute to the model; the finding does not show that laboratory values are unimportant for readmission in general.

The model's predictive performance was modest (AUC 0.633), and a logistic regression baseline performed almost identically (AUC 0.624), so the Random Forest did not offer a clear gain in predictive accuracy on this split. The SHAP analysis should therefore be read as describing the behaviour of the Random Forest, and a comparison of feature rankings across models was not performed. At the default threshold it identified half of the readmitted patients, but most flagged patients were not readmitted, so it is better suited to prioritising patients for review than to making individual decisions. From a practical standpoint, case-level SHAP explanations could allow a care team to see why a particular patient was flagged, for example because of repeated prior admissions or a transfer to a nursing facility, and to select an appropriate follow-up. Whether such explanations improve clinical decisions has not been tested here.

4.1 Limitations

Data limitations. The dataset covers encounters from 1999 to 2008, and clinical practice, coding systems, and readmission policies have changed since then, so the findings may not transfer to current patients. Three variables (`weight`, `payer_code`, and `medical_specialty`) had extensive missing values and were removed, and the key laboratory measures are largely missing in the dataset, so laboratory information had limited opportunity to contribute to the model. The dataset also records no social or environmental information, such as living situation, medication affordability, or access to follow-up care, which limits what the model can capture. Finally, readmission is recorded only if it occurred within the same hospital system, so some readmissions to other hospitals may be missed.

Cohort construction. Retaining only each patient's first encounter avoids repeated-patient dependence but discards later admissions, which may be the most clinically relevant for readmission risk, and it reduces the sample. Encounter order was approximated from encounter identifiers rather than admission dates, which the dataset does not provide.

Modelling limitations. Performance was evaluated on a single stratified train–test split, without cross-validation, so the estimates may vary with a different split. Hyperparameters were fixed rather than tuned, only one simple baseline model was compared, the difference in AUC between the two models (0.633 versus 0.624) was not tested for statistical significance, and classification was assessed at a single default threshold. Predictive performance was modest (AUC 0.633), and because only 9.0% of patients were readmitted, precision for the readmitted class was low (0.14). One-hot encoding of the diagnosis codes produced 2,291 sparse predictor columns, and grouping the codes into broader categories might improve both performance and interpretability. In addition, because the first category of each variable was dropped during encoding, patients with no A1C or glucose test share the all-zero indicator pattern with the reference result (A1C “>7”; glucose “>200”), so “not tested” is not distinguished from that result; an explicit “not tested” category would be a more complete encoding.

Interpretation limitations. SHAP values were computed on a random sample of 500 test patients rather than the full test set, so categories with few patients, such as discharge to a rehabilitation facility, are sparsely represented and their effects should be read cautiously. SHAP describes which features the model relies on, not what causes readmission, and several features may act as proxies for unmeasured factors such as illness severity or social support. Some findings, such as the lower predicted risk when diabetes is recorded as a secondary diagnosis, are not obviously clinical and may reflect coding practices.

Validation limitations. The explanations were not reviewed by clinicians, so their usefulness and trustworthiness in practice remain unvalidated, and the model was not tested on an external dataset.

5. Conclusion

A Random Forest classifier combined with SHAP can predict 30-day readmission in patients with diabetes with modest accuracy (AUC 0.633, compared with 0.624 for a logistic regression baseline) while providing case-level explanations. In this analysis, prior inpatient use, discharge to a skilled nursing facility, medication burden, and length of stay were among the most influential predictors, which suggests that a patient's recent utilisation history and discharge destination are useful signals for targeting transitional care. Future work should group diagnosis codes, use cross-validation and more recent data, compare feature rankings across models, incorporate social determinants of health such as access to pharmacies and the affordability of medication, and evaluate the explanations with clinicians.

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